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$$-1 - 2\sqrt{2}i = (x + yi)^2$$

$$\begin{cases} x^2 - y^2 &= -1 \\ 2xy &= -2\sqrt{2} \end{cases} \Leftrightarrow \begin{cases} x^2 - y^2 &= -1 \\ x^2 y^2 &= 2 \end{cases} \Leftrightarrow \begin{cases} x^2 - y^2 &= -1 \\ x^2(-y)^2 &= -2 \end{cases}$$

$$\Rightarrow \lambda^2 + \lambda - 2 = 0$$

$$D = 1^2 - 4 \cdot 1 \cdot (-2) = 9$$

$$\lambda_1 = \frac{-1 + 3}{2} = 1 = x^2 \Rightarrow x = \pm 1$$

$$\lambda_2 = \frac{-1 - 3}{2} = -2 = -y^2 \Rightarrow y = \pm\sqrt{2}$$

$$\Rightarrow z_1 = 1 - i\sqrt{2}$$

$$\Rightarrow z_2 = -1 + i\sqrt{2}$$

References